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Patients with Schizophrenia Have a Higher Risk of Psoriasis: A Systematic Review and Meta-analysis

Article type: Meta-analysis

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1. Introduction

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Recent studies have suggested that patients with schizophrenia may have a higher risk of developing several autoimmune diseases, suggesting the potential role of immune-mediated injury to the central nervous system (CNS) in the pathogenesis of schizophrenia (Bechter et al., 2013; Benros et al., 2014; Han et al., 2011; Kumar et al., 2013; Schmitt and Ford, 2010; Tu et al., 2017; Yu et al., 2017) although most of the studies could not demonstrate a significant association as both conditions are relatively uncommon in general population. Interestingly, a recent meta-analysis has demonstrated an increased risk of psoriasis among patients with Parkinson's disease (Ungprasert et al., 2016). Since the pathology of both schizophrenia and Parkinson's disease involve basal ganglia (Beal, 2001; Bechter et al., 2013), psoriasis maybe linked to schizophrenia as well. The purpose of the current systematic review and meta-analysis is to summarize all available evidence to investigate if schizophrenia is associated with a higher risk of psoriasis.

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2. Methods

2.1 Search strategy

Two investigators (P.U. and K.W.) independently searched published studies indexed in MEDLINE and EMBASE database from inception to May 2017 using the search strategy described in supplementary data. No language limitation was applied. A manual search of the bibliographies of selected original and review articles was also performed.

2.2 Inclusion criteria

Studies that investigated the risk of psoriasis among patients with schizophrenia are included. They could be either cohort or case-control studies. For cohort studies, they

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needed to have a cohort of patients with schizophrenia and a cohort of comparators without schizophrenia. They needed to report relative risk or hazard ratio with 95% confidence interval (CI) of incident psoriasis after the index date. For case-control studies, they needed to have patients with psoriasis as cases and subjects without psoriasis as controls. They needed to report odds ratio with 95% CI, comparing the odds of schizophrenia between cases and controls.

Study eligibility was independently determined by the two investigators (P.U. and W.K.). The third investigator (W.C.) reviewed all the included studied to ensure that the inclusion criteria were met and also served as the deciding vote when the first 2 investigators had different determinations of each study's eligibility. Quality of the included studies was appraised using the Newcastle-Ottawa quality assessment scale. Using this scale, each study was assessed in three main areas including (1) the selection of the study subjects (2) the comparability between the groups and (3) the ascertainment of the outcomes of interest (cohort studies) or the exposures of interest (case-control studies) (Stang, 2010).

2.3 Data extraction

A standardized data collection form was used to extract the following information: first author's name, title of the study, year of publication, year of study, country where the study was conducted, number of subjects, study population, diagnosis of schizophrenia, diagnosis of psoriasis, confounders that were adjusted and adjusted effect estimates with 95% CI. This data extraction was independently performed by the 2 investigators (P.U. and K.W.) to ensure the accuracy. Any discrepancies found in the case record forms were resolved by referring back to the original articles.

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2.4 Statistical analysis

Review Manager 5.3 software from the Cochrane Collaboration (London, United Kingdom) was used for the statistical analyses. Adjusted point estimates and standard errors from each study were pooled together using the generic inverse variance method of DerSimonian and Laird (DerSimonian and Laird, 1986). Given the high likelihood of between study variance due to different populations and study design, random-effect model, rather than fixed-effect model, was utilized. OR from case-control studies was used as an estimate to pool the effect estimates with relative risk from cohort study. Cochran's Q test was used to evaluate the statistical heterogeneity. The test was complemented with the I^2 statistic, which quantifies the proportion of total variation across studies that is due to heterogeneity rather than chance. A value of I^2 of 0% to 25% indicates insignificant heterogeneity, 26% to 50% indicates low heterogeneity, 51% to 75% indicates moderate heterogeneity and indicates 76% to 100% high heterogeneity (Higgins et al., 2003). Funnel plot and Egger's regression test were used to assess the presence of publication bias. Egger's regression test was performed using Comprehensive Meta-Analysis Software version 2.2 (Englewood, New Jersey, United States).

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3. **Results**

The search strategy yielded 294 potentially relevant articles (245 articles from EMBASE and 49 articles from Medline). After excluding 47 duplicated articles, 247 articles underwent title and abstract review. 227 articles were excluded as they were review articles, editorial, case reports, correspondence, animal studies or interventional studies,

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leaving 20 articles for full-length article review. Five articles were excluded at this stage as they reported risk of schizophrenia among patients with psoriasis (not risk of psoriasis among patients with schizophrenia) whereas 9 articles were excluded because they did not report the outcome of interest. Therefore, 2 retrospective cohort studies and 4 case-control studies with approximately 6.1 million participants met the pre-defined eligibility criteria and were included in the meta-analysis (Benros et al., 2014; Han et al., 2011; Kumar et al., 2013; Schmitt and Ford, 2010; Tu et al., 2017; Yu et al., 2017). Figure 1 outlines the literature review and selection process. The main characteristics and Newcastle-Ottawa scores of the included studies are described in table 1.

The pooled relative risk of psoriasis among patients with schizophrenia versus non-schizophrenia subjects was 1.83 (95% CI, 1.36-2.47). The statistical heterogeneity of this meta-analysis was moderate with an I^2 of 73%. Subgroup analysis by study design revealed pooled relative risk of 1.48 (95% CI, 0.82 – 2.64; I^2 68%) for case-control studies and pooled relative risk of 2.22 (95% CI, 1.95 – 2.52; I^2 0%) for cohort studies. Figure 2 demonstrates the forest plot of this meta-analysis.

Sensitivity analyses were conducted by excluding the study by Tu et al. and Yu et al. from the overall analysis (one study at a time) as both studies used the Taiwan National Health Institute Database and may have included the same patients. The results of the sensitivity analyses were not significantly different from the overall analysis. The pooled relative risk was 1.71 (95% CI 1.20 – 2.43) after exclusion of the study by Tu et al. and the pooled relative risk was 1.65 (95% CI 1.08 – 2.51) after exclusion of the study by Yu et al.

3.1 Evaluation for publication bias

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Visualization of funnel plot (figure 3) did not provide suggestive evidence of publication bias in favor of positive studies as the graph was relatively symmetric. In addition, there was no evidence of publication bias based on Egger's regression test (p = 0.49).

4. Discussion

The current study is the first systematic review and meta-analysis to assess the risk of psoriasis in patients with schizophrenia. We are able to demonstrate a 1.83-fold increased risk of psoriasis among patients with schizophrenia compared with subjects without schizophrenia. This increased risk may warrant more attention from psychiatrists and other healthcare providers on skin health of patients with schizophrenia. Evaluation by dermatologist may be necessary if patients develop skin eruptions for early diagnosis and treatment of psoriasis.

The pathogenesis of this association is not known but there are some potential explanations. Immune dysregulation could be the shared underlying etiology for both conditions. Psoriasis is a chronic immune-mediated inflammatory disorder with T helper 17 (Th17) cells and tumor necrosis factor (TNF)-alpha being the major players for its pathogenesis. According to the mild encephalitis hypothesis, which is supported by immunologic, cerebrospinal fluid and imaging studies, inflammation of the CNS may play a pivotal role in the development of schizophrenia in a significant subgroup of patients (Bechter et al., 2013; Debnath and Berk, 2014; Ding et al., 2014; Na et al., 2014; Riedmuller and Muller, 2017; Tu et al., 2017). These studies also suggest a significant role of Th17 cells

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for the initiation and progression of the inflammation in the CNS (Debnath and Berk, 2014; Ding et al., 2014; Na et al., 2014; Riedmuller and Muller, 2017). Thus, it is possible that dysregulated Th17 cells can predispose patients to both conditions.

A recent genome-wide association study has demonstrated a strong association between schizophrenia and immune system (Ripke et al., 2014). In this study, schizophrenia-associated loci were found to be overly expressed in tissues with important immune function, especially, from B-cell lineages required for intact adaptive immunity which corresponds to current knowledge of immune dysregulation underlying pathogenesis of psoriasis. Another genome-wide association study found candidate immune-related genes encoding alternative, non-immune related proteins in the brain (Pouget et al., 2016).

It is also possible the apparent association between the 2 disorders is the result of confounders. For example, it is known that patients with schizophrenia have a higher prevalence of substance abuse including smoking and alcohol (Regier et al., 1990) and both are also risk factors for psoriasis (Lonnberg et al., 2016; Farkas and Kemeny, 2013; Adamzik et al., 2013).

It is unlikely that non-specific inflammation would be the key link between the 2 conditions as the risk for individual autoimmune disease is often not increased among patients with schizophrenia (Benros et al., 2014). In fact, the negative relationship between schizophrenia and rheumatoid arthritis, one of the most common chronic inflammatory disorders, has been observed for decades (Euesden et al., 2015). Shared specific inflammatory pathways or factors unrelated to inflammation (as previously discussed) are, therefore, the more plausible explanations.

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The major strength of this study was that of the nature of systematic review and meta-analysis that comprehensively combined all evidence together. The primary studies included in this analysis were also of high quality as reflected by the high Newcastle-Ottawa Scores. Nonetheless, the study had some limitations and the interpretation of the results should be done with caution. The major limitation is that all of the included studies (except for the study by Kumar et al.) were conducted using electronic medical record database and relied on diagnostic codes to identify schizophrenia and psoriasis. Diagnostic codes in electronic medical record database are known to have low sensitivity and specificity which might jeopardize the validity of the effect estimates from the primary studies. Another limitation is the statistical heterogeneity that was moderate-to-high in this analysis. Difference in study design could be responsible to this as subgroup analysis per study design revealed lower I^2 for both subgroups (compared with the overall analysis). It should be noted that the I^2 of cohort study subgroup was 0% which may reflect the similarity between the 2 studies that were both retrospective cohort studies using national databases. On the other hand, the I^2 of case-control study subgroup remained relatively high which may reflect the nature of case-control study that is more prone to bias and, thus, higher likelihood of scattered results.

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5. Conclusion

In conclusion, this systematic review and meta-analysis of cohort and case-control studies demonstrated a significantly increased risk of psoriasis among patients with schizophrenia with 83% excess risk.

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Conflict of interest

None

Figure legend

Figure 1: Literature review and selection process.

Figure 2: Forest plot of all included studies. The size of the box used for each study represents the weight of that study used for calculation of the pooled effect. The diamond at the bottom of this forest plot represents the pooled relative risk with 95 % confidence interval (the center of the diamond represents the pooled relative risk and the lateral lips of diamond represent its 95 % confidence interval).

Figure 3: Funnel plot. Each circle represents one study. X-axis: effect estimate (relative risk) of each study. The dotted vertical line represents the pooled relative risk. Y-axis: standard error of log relative risk which was used to assign the weight of each study for the pooled analysis.

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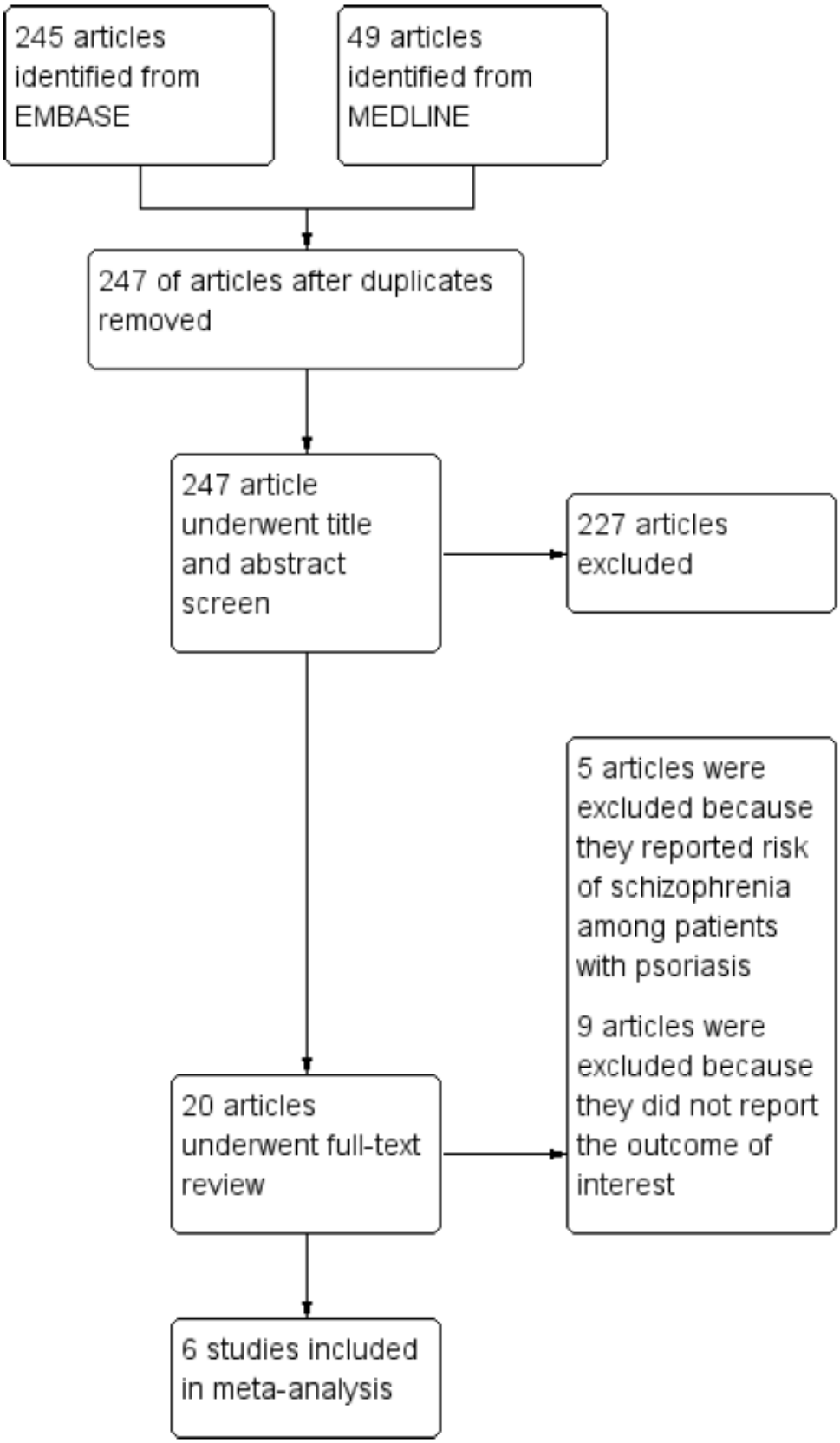
245 articles identified from EMBASE
49 articles identified from MEDLINE
247 of articles after duplicates removed
247 article underwent title and abstract screen
227 articles excluded
5 articles were excluded because they reported risk of schizophrenia among patients with psoriasis
9 articles were excluded because they did not report the outcome of interest
20 articles underwent full-text review
6 studies included in meta-analysis

Fig. 1

Study or Subgroup	log[Risk Ratio]	SE	Weight	Risk Ratio IV, Random, 95% CI	Year
Schmitt et al.	-0.0723	0.2182	18.2%	0.93 [0.61, 1.43]	2010
Han et al.	0.2624	0.3192	12.8%	1.30 [0.70, 2.43]	2011
Benros et al	0.7561	0.0776	27.0%	2.13 [1.83, 2.48]	2014
Kumar et al.	1.0986	1.6131	0.9%	3.00 [0.13, 70.83]	2015
Tu et al.	0.9123	0.2418	16.8%	2.49 [1.55, 4.00]	2017
Yu et al.	0.8961	0.1217	24.5%	2.45 [1.93, 3.11]	2017
Total (95% CI)			**100.0%**	**1.83 [1.36, 2.47]**	

Heterogeneity: Tau² = 0.08; Chi² = 18.24, df = 5 (P = 0.003); I² = 73%
Test for overall effect: Z = 3.97 (P < 0.0001)

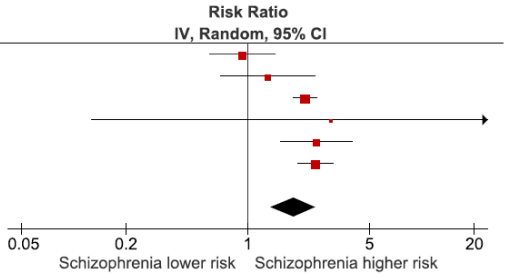

Fig. 2

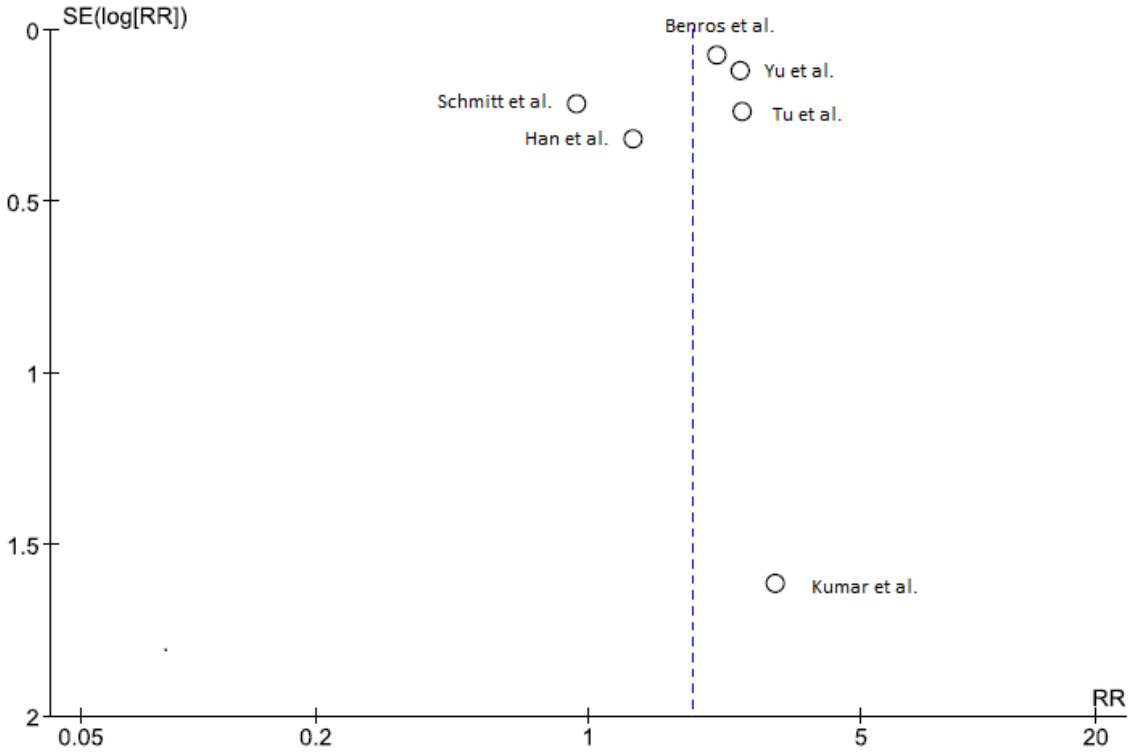

Fig. 3

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Table 1: Main characteristics of the studies included in this meta-analysis

	Schmitt et al. [5]	Han et al. [6]	Benros et al. [7]	Kumar et al. [8]	Tu et al.[9]	Yu et al. [10]
Country	Germany	United States	Denmark	India	Taiwan	Taiwan
Study design	Case-control study	Case-control study	Retrospective cohort study	Case-control study	Case-control study	Retrospective cohort study
Year of publication	2010	2011	2014	2015	2017	2017
Number of participants	6,294 (3,147 cases / 3,147 controls)	39,855 (7,971 cases / 31,884 controls)	3.77 million (39,364 cases of schizophrenia)	60 (30 cases / 30 controls)	21,592 (10,796 cases / 10,796 controls)	24,900 (4,980 cases / 19,920 controls)
Subjects	Cases: patients with psoriasis were identified from the GKV-database Saxony between 2003 and 2004. Controls: sex and age-matched controls without psoriasis were randomly selected from the	Cases: patients with psoriasis were identified from an insurance claim database (PharMetrics Patient Centric Database) between 2003 and 2004. Controls: sex and age-matched controls without psoriasis	Cases: patients with schizophrenia were identified from the Danish Civil Registration System between from 1987 and 2010. Comparators: the rest of the patients without diagnosis of schizophrenia in the register	Cases: patients with psoriasis were recruited from the study center Controls: controls without psoriasis were recruited from the same hospital.	Cases: patients with psoriasis were identified from the database of National Health Institute Research Database in 2010. Controls: sex and age-matched controls without psoriasis were randomly selected	Cases: patients with schizophrenia were identified from the database of National Health Institute Research Database between 1996 and 2010 Comparators: sex and age-matched controls without psoriasis were

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	same database.	were randomly selected from the same database.	served as comparators.		from the same database.	randomly selected from the same database.
Diagnosis of psoriasis	Presence of diagnostic codes of psoriasis in the database	Presence of diagnostic codes of psoriasis in the database	Presence of diagnostic codes of psoriasis in the database	Interview and physical exam by a dermatologist	Presence of diagnostic codes of psoriasis in the database	Presence of diagnostic codes of psoriasis in the database
Diagnosis of schizophrenia	Presence of diagnostic codes of schizophrenia in the database	Presence of diagnostic codes of schizophrenia in the database	Presence of diagnostic codes of schizophrenia in the database	Interview and physical exam by a psychiatrist	Presence of diagnostic codes of schizophrenia in the database	Presence of diagnostic codes of schizophrenia in the database
Confounder adjustment	Age, sex, urbanization, socioeconomic status and family history of schizophrenia	Age, sex, region and time-in-plan	Age, sex and calendar year	Age, sex, urbanization, socioeconomic status and geographic location	Age, sex and comorbidities	Age, sex and comorbidities
Quality assessment (Newcastle-Ottawa Score)	Selection: 3 Comparability: 2 Outcome: 3	Selection: 3 Comparability: 2 Outcome: 3	Selection: 3 Comparability: 1 Outcome: 3	Selection: 2 Comparability: 0 Outcome: 3	Selection: 3 Comparability: 1 Outcome: 3	Selection: 3 Comparability: 2 Outcome: 3

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Highlight

- Epidemiologic studies demonstrated an increased risk of psoriasis among patients with schizophrenia. However, the results are inconclusive.
- This meta-analysis of cohort and case-control studies demonstrated a significantly increased risk of psoriasis among patients with schizophrenia with 83% excess risk.
- The pathogenesis of this association remains unknown. Immune dysregulation, the susceptibility genes among psoriasis and schizophrenia may play a role in the pathogenesis.
- This study will raise awareness of the increased risk of psoriasis among patients with schizophrenia.

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